PREP-RASHAN

A Unit of Prep-Rashan Tutorial

DAILY CURRENT AFFAIRS - RASHAN

Comprehensive Compilation for UPSC Civil Services Examination

4 November 2026

General Studies Paper I • Paper II • Paper III • Paper IV

Keywords Defined • PYQ (Prelims & Mains) • Practice Questions (Prelims & Mains)

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GENERAL STUDIES PAPER I

[GS-I] Indian Heritage & Culture, History, Geography of the World and Society

GS-I | Indian Society – Urbanisation

1. Why India Needs a National Urbanisation Policy

Syllabus: *GS-I: Urbanisation – problems and remedies. GS-II: Local self-government (74th Amendment); federalism. GS-III: Infrastructure.*

CONTEXT

India's towns are **growing far beyond their municipal boundaries**, but the **next Census** will still use the **2011 rules** to decide what counts as 'urban'. Experts argue for a **National Urbanisation Policy (NUP)** that brings **classification, planning, finance and governance** under one framework.

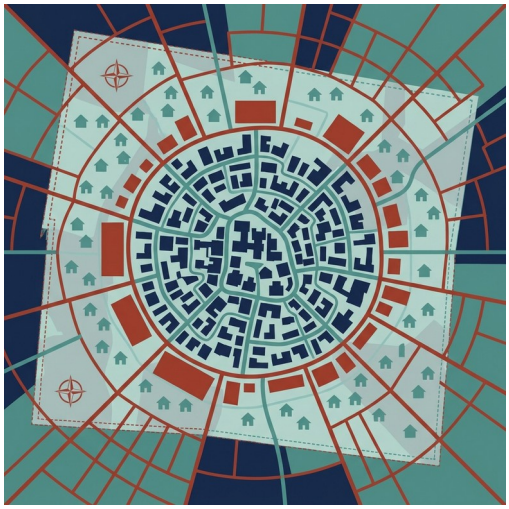


Illustration: a dense city core spilling into peri-urban villages, with official boundary lines that no longer match the built-up area.

KEYWORDS DEFINED

Keyword	Meaning
Statutory town	A place declared urban by the **State government by notification**, with a **municipality, municipal corporation, nagar panchayat or cantonment board**.
Census town	A place still run by a **gram panchayat** but meeting three tests: **population 5,000+**, **density 400+ per sq km**, and **75%+ of male main workers outside farming**.
Peri-urban area	The **fringe around a city** where land use, jobs and commuting are urban but **administration is still rural**.
Functional urban region	An area defined by **how people live and work** (commuting, markets, services) rather than by **administrative lines**.
Twelfth Schedule	The list of **18 functions** (planning, water, roads, sanitation, etc.) that States **may** devolve to municipalities under the **74th Amendment**.

How India counts 'urban' today

- **Two routes: statutory towns** (by State notification) and **census towns** (by the three population and workforce tests).
- **Article 243Q(2):** the **Governor** notifies urban areas, so **criteria differ between States**.
- **Rural–urban binary:** every settlement is either rural or urban – the **in-between belt is ignored**.
- **Broad size classes:** 'Class I' covers every town of **1 lakh or more**, lumping very different places together.
- **Why the label matters:** it decides access to **master plans, municipal finance, Finance Commission grants, building rules and schemes**.

The scale of the gap

- **Census 2011:** about **377 million** urban residents in roughly **8,000** towns; the **UN** expects **720 million+** by **2050**.
- **Satellite view:** the **Economic Survey 2025–26** cited estimates that India was already **63% urban in 2015** – far above the official count.
- **Fast fringe:** peri-urban areas grow about **4.15% a year**, against **2.9%** inside city limits; States have notified about **1,000 more statutory towns** since 2011.
- **Example cited:** the **Bhiwandi** logistics belt near Mumbai spans **60 villages and 9 census towns** with over **90% non-farm jobs**, yet is run by **village panchayats**.

Problems that follow

- **Governance gap:** panchayats handle **land use, waste, water and air** without the **mandate, engineers or money** for urban densities.
- **No plans:** about **52% of statutory towns** and **76% of census towns** have **no master plan**; **nine States** have not updated planning laws since the 74th Amendment, and the **model law dates to 1985**.
- **Weak finances:** municipal revenue is only about **1% of GDP** (Brazil **7.4%**, South Africa **6%**); **property tax** yields just **0.15% of GDP**.
- **Partial devolution: Article 243W** is **permissive**, so notification does not automatically pass on all **18 Twelfth Schedule functions**.
- **Wrong scale for pollution:** about **65% of Delhi's PM2.5** comes from **outside the city**, so city-only action falls short.

What a National Urbanisation Policy would do

- **Common classification rules** using **satellite and economic data**, reviewed **between Censuses**.
- **Modern planning laws** with **regional plans** and **peri-urban zoning**.
- **Predictable finance:** stronger **own revenue** (property tax), **formula-based untied grants** and help to **prepare bankable projects**.
- **A National Urbanisation Council** linking the **Inter-State Council, NITI Aayog** and local-government bodies; active **District and Metropolitan Planning Committees** (**Articles 243ZD and 243ZE**).
- **Value addition:** the **Union Budget 2026** set aside **₹5,000 crore for each City Economic Region**; a **₹10,000-crore 'urbanisation premium'** linked to the **16th Finance Commission** is meant to reward States that merge fringe villages into towns *(as cited)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about the **74th Amendment**, the **Twelfth Schedule**, **census town criteria** and **Metropolitan/District Planning Committees**.

PYQ – MAINS

- **Pattern:** Mains asks about **urban planning failures**, **municipal finance** and **smart, inclusive cities**.

FQ – PRELIMS (Possible Question)

- To be classified as a Census Town in India, a settlement must have: 1. a population of at least 5,000; 2. a density of at least 400 persons per sq km; 3. a notified municipality. Which of these are required? **(Ans: 1 and 2 only)**

FQ – MAINS (Possible Question)

- "India is more urban than its Census says." Examine the gaps in urban classification and suggest the elements of a National Urbanisation Policy. (250 words)

GS-I | Geography – Places in News (Mapping)

2. Qeshm Island: Iran's Largest Island at the Strait of Hormuz

Syllabus: *GS-I: Location of important geographical features (mapping). GS-II: Effect of policies of other countries on India's interests (energy, shipping).*

CONTEXT

The **United States** carried out **limited strikes on Qeshm Island** in the **Strait of Hormuz**, saying it was targeting **underground missile sites** used to threaten **merchant shipping**.



Satellite view of Qeshm Island, the long arrow-shaped island in the Strait of Hormuz off Iran's coast.
Photo: NASA image using data provided courtesy of the University of Maryland's Global Land Cover Facility, Wikimedia Commons, Public domain

KEYWORDS DEFINED

Keyword	Meaning
Strait of Hormuz	The narrow sea passage linking the **Persian Gulf** with the **Gulf of Oman** and the **Arabian Sea** – a key route for **oil and LNG**.
Clarence (Khuran) Strait	The narrow channel that **separates Qeshm from mainland Iran**.
Free trade zone	An area with **relaxed customs and tax rules** to attract trade and investment.
Geopark	A protected area with **outstanding geology** managed for conservation, education and tourism (UNESCO Global Geoparks).
A2/AD	**Anti-access/area denial** – using missiles, mines and fast boats to **keep an enemy's ships away** from a zone.

Location

- The **largest island in the Persian Gulf**, in Iran's **Hormozgan province**, lying across the **Strait of Hormuz**.
- Separated from the port city of **Bandar Abbas** and the mainland by the narrow **Clarence (Khuran) Strait**.
- Lies roughly **60 km north of Oman's Musandam peninsula**, the other shore of the strait.

Key features

- **Size:** about **1,490–1,500 sq km**, nearly **135 km long**; highest point **Mount Bukhun (~370 m)**.
- **Namakdan Salt Cave** – about **6.6 km long**, among the world's longest salt caves – in the **Qeshm Island UNESCO Global Geopark**.

- **Hara mangrove forests** on the north-west coast shelter fish and **migratory birds**.
- An **Iranian free trade zone** and reportedly a base for **naval tunnels, anti-ship missiles and fast attack craft**.

Why the strikes matter

- **Oil route at risk:** tension near Hormuz raises fears of **shipping disruption** and **oil price spikes**.
- **Weakening A2/AD:** hitting missile sites may **reduce Iran's ability to threaten shipping**, but **retaliation** could widen the conflict.
- **For India:** a large share of India's **crude, LPG and LNG** comes through Hormuz; **Indian sailors** work on Gulf routes; **Chabahar port** lies just outside the strait in the Gulf of Oman.
- **Value addition:** other islands near the strait include **Hormuz and Larak**; **Abu Musa and the two Tunbs** are held by Iran but **claimed by the UAE**.

PYQ – PRELIMS

- **Pattern:** Prelims mapping asks which **straits, islands and gulfs** lie between given countries (Hormuz, Bab-el-Mandeb, Malacca).

PYQ – MAINS

- **Pattern:** Mains asks about **West Asian instability** and its effect on **India's energy security and diaspora**.

FQ – PRELIMS (Possible Question)

- Qeshm Island, recently in the news, lies in: (a) the Red Sea (b) the Strait of Hormuz (c) the Caspian Sea (d) the Gulf of Aden **(Ans: b)**

FQ – MAINS (Possible Question)

- "Chokepoints decide energy security." Discuss the importance of the Strait of Hormuz for India and the steps India has taken to reduce its risks. (150 words)

GS-I | Indian Culture – Traditional Knowledge Systems **UPDATE**

3. IJTK at 25: A Journal That Guards India's Traditional Knowledge

Syllabus: *GS-I: Indian culture – knowledge traditions. GS-III: Science and technology; intellectual property rights and biopiracy.*

WHAT WE KNOW TILL NOW

- **CDTK-2026**, the **third international conference** on **communicating and spreading traditional knowledge**, was held at **CSIR-NPL, New Delhi**.
- It ran under **SVASTIK** (**Scientifically Validated Societal Traditional Knowledge**), a **CSIR-NIScPR** initiative of **2021** that promotes only **scientifically tested** traditional practices.
- **TKDL** helps patent offices **reject claims on known knowledge** – as in the **turmeric and neem** cases.

WHAT TODAY ADDS

- The conference **concluded** and marked **25 years** of the **Indian Journal of Traditional Knowledge (IJTK)**.
- What the journal is, how it is **indexed**, and how it serves as **prior art** against wrongful patents.

CONTEXT

CSIR-NIScPR closed **CDTK-2026**, which also celebrated the **25th year** of the **Indian Journal of Traditional Knowledge (IJTK)** – a research journal that records and **scientifically tests** India's traditional knowledge.



Illustration: an open research journal beside herbs, a mortar and pestle, a microscope and a laptop – documenting and testing traditional knowledge.

KEYWORDS DEFINED

Keyword	Meaning
Peer review	Checking of a research paper by **independent experts** before it is published.
Diamond open access	A journal that is **free to read and free to publish in** – no fees for readers or authors.
Prior art	Any **earlier public evidence** of an idea; it can be used to **reject a patent** claimed as new.

Keyword	Meaning
Reverse pharmacology	Starting from a **traditional remedy known to work** and then studying it with modern science to build a medicine.
Ethnobotany	The study of **how communities use plants** for food, medicine and rituals.

About IJTK

- A **peer-reviewed, interdisciplinary quarterly** that documents and **scientifically validates** traditional knowledge, indigenous practices and bio-resources.
- **Started in 2002** by **CSIR** through the then **NISCAIR** – now **CSIR-NIScPR** – and is in its **25th year** in 2026.
- **Aim:** document, test and preserve TK in **health care, biodiversity and rural technology**, and bring it into **evidence-based research**.

Key features

- **Scope: AYUSH** systems, **ethnobotany, ethnomedicine, ethnopharmacology** and traditional remedies.
- **Prior-art status:** listed in the **minimum documentation** that patent examiners consult under **WIPO's Patent Cooperation Treaty (PCT)** – a shield against **biopiracy**.
- **Indexed** in **Scopus**, the **Science Citation Index Expanded** and **Journal Citation Reports**.
- **Diamond open access:** no reading or publishing charges; **single-blind peer review** with stress on **community consent and benefit-sharing**.

Why it matters

- Works alongside the **Traditional Knowledge Digital Library (TKDL)** to **block wrongful patents** on Indian knowledge.
- Supports **reverse pharmacology** – a faster, cheaper route to **affordable, standardised medicines**.
- **Value addition:** the **Biological Diversity Act, 2002** and the **Nagoya Protocol (2010)** require **access and benefit-sharing** when bio-resources and related knowledge are used commercially.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **TKDL**, **CSIR institutions**, **GI tags**, **biopiracy** and the **Nagoya Protocol**.

PYQ – MAINS

- **Pattern:** Mains asks how India can **protect traditional knowledge** and turn it into **evidence-based innovation**.

FQ – PRELIMS (Possible Question)

- Consider: 1. The Indian Journal of Traditional Knowledge is published by CSIR-NIScPR. 2. It charges authors a fee for open-access publication. 3. It is part of the PCT minimum documentation used by patent examiners. Which are correct? **(Ans: 1 and 3 only)**

FQ – MAINS (Possible Question)

- "Documentation is the first defence against biopiracy." Discuss the role of IJTK and TKDL in protecting India's traditional knowledge. (150 words)

GENERAL STUDIES PAPER II

[GS-II] Governance, Constitution, Polity, Social Justice & International Relations

GS-II | International Relations – India–Bangladesh; Water Sharing

4. Ganga Water Treaty: Renewal Talks Before the 2026 Expiry

Syllabus: *GS-II: India and its neighbourhood; bilateral agreements. GS-I: Drainage systems – the Ganga. GS-III: Water resources.*

CONTEXT

With the **Ganga Water Treaty** due to **expire in December 2026**, India said that a **structured mechanism already exists** to deal with the treaty's requirements, while **Bangladesh is pressing for renewal** on better terms.



The road on top of the Farakka Barrage, West Bengal, where Ganga waters are shared with Bangladesh in the dry season.
Photo: Pinakpani, Wikimedia Commons, CC BY-SA 4.0

KEYWORDS DEFINED

Keyword	Meaning
Lean season	The **dry months (1 January to 31 May)** when river flow is lowest and sharing matters most.
Cusec	**One cubic foot of water per second** – the unit used to measure river flow.
Farakka Barrage	A barrage on the Ganga in **Murshidabad, West Bengal**, built to **divert water into the Hooghly** to keep **Kolkata port** navigable.
Guarantee clause	A promise in the **1977 agreement** that Bangladesh would get at least **80% of its share** even in very low flows – **not in the 1996 treaty**.
Environmental flow	The **minimum water** a river needs to keep its **ecosystem and people healthy**.

About the treaty

- A **30-year India–Bangladesh agreement** signed on **12 December 1996** by Prime Ministers **H.D. Deve Gowda** and **Sheikh Hasina**.
- It shares the Ganga's water at the **Farakka Barrage** during the **lean season (1 January – 31 May)**.
- **Sharing formula:** if flow is **70,000 cusecs or less**, it is **split equally**; between **70,000 and 75,000 cusecs**, Bangladesh gets **35,000** and India the rest.
- **Assured share:** each side gets at least **35,000 cusecs** in **alternate ten-day periods** between **11 March and 10 May**.
- **Emergency clause (Article II):** if flow at Farakka falls **below 50,000 cusecs** in any ten-day period, both must **consult at once** to adjust shares.

Bangladesh's demands

- **Renew the treaty before 12 December 2026.**
- **Restore the guarantee clause** of the **1977 agreement** (at least **80%** of its share in very low flows).
- Add **environmental flows** and **climate change**, which the 1996 text does not cover.
- Some experts suggest bringing in **Nepal** to make it **trilateral**.

About the Ganga (geography)

- Rises from the **Gangotri glacier** (Uttarkashi, Uttarakhand) as the **Bhagirathi**; becomes the **Ganga at Devprayag** after meeting the **Alaknanda**.
- About **2,525 km** long; the Indian basin covers roughly **26%** of India's area across **Uttarakhand, UP, Bihar, Jharkhand and West Bengal**.
- Called the **Padma** in Bangladesh; main tributaries include the **Yamuna, Son, Ramganga, Ghaghara, Gandak, Kosi and Mahananda**.
- **Value addition:** other India–Bangladesh water deals – the **Kushiyara MoU (2022)** and the **Feni MoU (2019)** for drinking water to **Sabroom, Tripura**; the **Teesta** deal is still pending.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **river-sharing treaties** (Indus, Ganga), **barrages** and **tributaries** of the Ganga.

PYQ – MAINS

- **Pattern:** Mains asks about **trans-boundary rivers** and **India–Bangladesh relations**.

FQ – PRELIMS (Possible Question)

- Under the Ganga Water Treaty, 1996, water is shared at: (a) Hardinge Bridge (b) Farakka Barrage (c) Bhimnagar Barrage (d) Tehri Dam **(Ans: b)**

FQ – MAINS (Possible Question)

- "Renewing the Ganga Water Treaty is a test of India's neighbourhood-first policy." Discuss the issues involved and suggest a balanced approach. (250 words)

GS-II | International Relations – Disarmament & the UN

5. UNGA Vote on Regional Arms Control: Why India Said No

Syllabus: *GS-II: Important international institutions – the UN General Assembly; India's foreign policy and security interests.*

CONTEXT

India and Israel voted against a **draft resolution on regional and subregional conventional arms control**, sponsored by **Pakistan and Belarus**, in the **UN General Assembly's First Committee**.

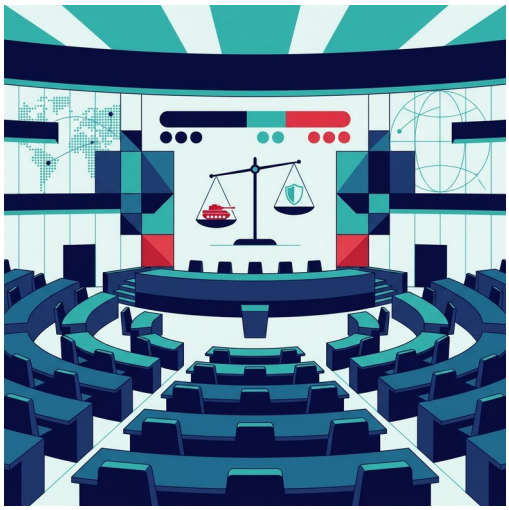


Illustration: a UN-style assembly hall with a voting board and a balance weighing tanks and shields.

KEYWORDS DEFINED

Keyword	Meaning
First Committee	The UNGA committee on **Disarmament and International Security**.
Conventional arms	**Non-nuclear weapons** – tanks, artillery, aircraft, ships, small arms.
Confidence-building measures (CBMs)	Steps like **hotlines, prior notice of exercises and data exchange** that reduce suspicion between states.
UN Disarmament Commission (UNDC)	A UN body that makes **recommendations and guidelines** on disarmament; it works by **consensus**.
Military equilibrium	A **balance of forces** between rivals, often argued for at the **lowest possible level**.

What the resolution says

- A **yearly resolution** in the First Committee under **general and complete disarmament**, tabled by **Pakistan** (with **Belarus** this time).
- Pushes **regional and subregional** deals to limit **conventional forces, weapons and defence spending** at the **lowest level**.
- Gives a **special responsibility** to the **militarily stronger state** in each region to start cuts.
- Asks the **UNDC** to draft **guidelines** on regional conventional arms control and refers to **South Asia** in particular.

India's stand

- **Threats are not only regional:** India faces challenges from **beyond South Asia** (e.g., **China**, the **Indian Ocean**), so a **regional balance** with one neighbour makes no sense.
- **Global, non-discriminatory rules** are better than **region-specific ceilings**.
- **Objects to the South Asia paragraph**, which would restrict India's defence needs.
- **No new guidelines needed:** the UNDC already adopted texts in **1993** (regional disarmament), **1999** (conventional arms control) and **2017** (practical CBMs).

Significance

- Shows India's preference for **independent defence planning** and **strategic autonomy**.
- **Israel's** no vote reflects a similar worry about **multi-front threats**.
- **Value addition:** India supports the **UN Register of Conventional Arms** and is a member of the **Wassenaar Arrangement (2017)**, **MTCR (2016)** and **Australia Group (2018)** – but **not the NSG**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **export-control regimes** (MTCR, Wassenaar, Australia Group, NSG) and **UN bodies**.

PYQ – MAINS

- **Pattern:** Mains asks about India's position on **disarmament, non-proliferation** and **multilateral forums**.

FQ – PRELIMS (Possible Question)

- Consider: 1. The First Committee of the UNGA deals with disarmament and international security. 2. India is a member of the Wassenaar Arrangement. 3. India supported the draft resolution on regional conventional arms control. Which are correct? **(Ans: 1 and 2 only)**

FQ – MAINS (Possible Question)

- "India's security calculus cannot be confined to South Asia." Explain this in the context of India's vote on regional conventional arms control at the UN. (150 words)

GS-II | International Relations – India–US Defence Cooperation

6. Tiger Triumph 2026: India–US Tri-Service Drill for Disaster Relief

Syllabus: *GS-II: Bilateral relations – India–US. GS-III: Security forces; disaster management.*

CONTEXT

The **5th edition** of **Tiger Triumph**, the **India–US tri-service exercise** on **Humanitarian Assistance and Disaster Relief (HADR)**, opened at the **Maritime Warfare Centre, Visakhapatnam**, with the **NDRF** taking part for the first time.



Ships and aircraft of India and the US sailing and flying in formation during an earlier Tiger Triumph exercise.
Photo: U.S. Marine Corps photo by Cpl. Aidan Hekker, Wikimedia Commons, Public domain

KEYWORDS DEFINED

Keyword	Meaning
HADR	**Humanitarian Assistance and Disaster Relief** – using forces to help people after **cyclones, floods and earthquakes**.
Tri-service	Involving all three wings – **Army, Navy and Air Force**.
Interoperability	The ability of two militaries to **work together smoothly** – compatible plans, communications and procedures.
Tabletop exercise (TTX)	A **discussion-based drill** where planners work through a scenario on maps and screens instead of in the field.
NDRF	**National Disaster Response Force** – India's specialised disaster force under the **Disaster Management Act, 2005**.

About the exercise

- India's **only bilateral tri-service exercise with the US**; the other tri-service exercise India holds with a single partner is **INDRA** with **Russia**.
- **First held in 2019**; **2026 is the 5th edition**, **hosted by India** at the **Maritime Warfare Centre (Eastern Naval Command), Visakhapatnam**.
- **Aim:** better **joint planning, coordination and interoperability** for large-scale **HADR** across the **Indo-Pacific**.

What is new in 2026

- **Tabletop-heavy:** focus on **staff planning, crisis management and multi-agency coordination** rather than large field landings.
- **NDRF's first participation** – stronger **civil–military** coordination.
- **All domains:** Indian **Army, Navy, Air Force and Special Forces**; US **Army, Navy, Air Force and Marine Corps**, plus **space, cyber, medical and explosive-ordnance** specialists.
- **Integrated command and control** across **land, sea, air, cyber and space**.

Significance

- Turns the **Major Defence Partnership** and agreements like **LEMOA, COMCASA and BECA** into **practical joint work**.
- India is often the **first responder** in the region (e.g., **Operation Brahma**, Myanmar earthquake, 2025) – shared drills make **joint relief** faster.
- **Value addition:** other India–US exercises – **Yudh Abhyas** (Army), **Malabar** (navies, with Japan and Australia), **Cope India** (Air Force), **Vajra Prahar** (special forces).

PYQ – PRELIMS

- **Pattern:** Prelims asks to **match exercises with partner countries** and services.

PYQ – MAINS

- **Pattern:** Mains asks about **India–US defence ties** and India's role as a **first responder** in the Indo-Pacific.

FQ – PRELIMS (Possible Question)

- Tiger Triumph, recently in the news, is: (a) an India–Russia army exercise (b) an India–US tri-service HADR exercise (c) a Quad naval exercise (d) an India–Japan air exercise **(Ans: b)**

FQ – MAINS (Possible Question)

- "Disaster relief is the softest face of hard power." Discuss with reference to India's HADR cooperation with the US and its neighbours. (150 words)

GENERAL STUDIES PAPER III

[GS-III] Technology, Economic Development, Bio-diversity, Environment, Security & Disaster Management

GS-III | Indian Economy – Critical Minerals & Strategic Autonomy

7. Critical Minerals: Why a Stockpile Needs Processing Power

Syllabus: *GS-III: Indian economy – mobilisation of resources, industrial policy; infrastructure (energy); security of supply chains.*

CONTEXT

India is preparing a **strategic stockpile of critical minerals** to protect industry from **supply shocks**. The argument now is that **raw stocks alone are not enough** – India also needs the ability to **process them into usable industrial inputs**.

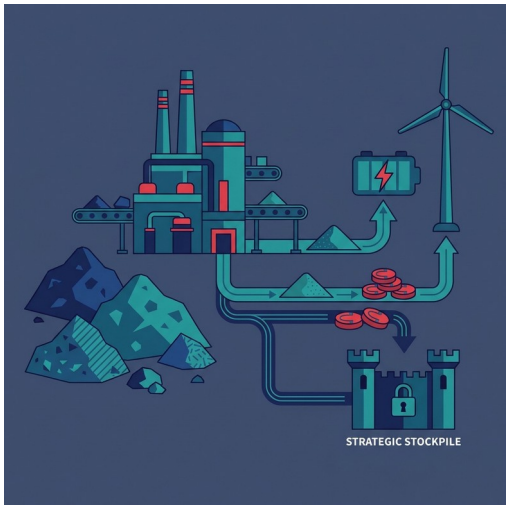


Illustration: raw ore passing through a processing plant into refined powders and magnets for batteries and wind turbines, with a secure stockpile warehouse.

KEYWORDS DEFINED

Keyword	Meaning
Critical minerals	**Non-fuel minerals** that are **vital for the economy and security** and face **supply risk** because extraction or processing is concentrated in a few places.
Strategic stockpile	A **government-held reserve** of key materials to keep industries running during **shortages or disruptions**.
Rare-earth permanent magnet (REPM)	Strong magnets made from **rare earths like neodymium**; used in **EV motors, wind turbines, electronics and defence**.
Midstream processing	The stage that turns **ore into refined chemicals, oxides and metals** – where most global capacity sits in **China**.
Urban mining	Recovering metals from **used batteries, electronics and scrap** – i.e., recycling.

Why stockpile

- **Supply security** against global disruptions and **geopolitical pressure**.
- **Industrial continuity** – factories keep running during a temporary shortage.
- **Strategic autonomy** in **defence, energy and electronics**, where a single missing input can halt production.

Limits of a stockpile

- **Ore is not a product:** owning raw minerals does not mean being able to turn them into **specific grades and chemical forms** that industry needs.
- **Technology gap:** processing needs **specialised equipment, know-how and infrastructure**, often foreign.
- **Small part, big damage:** a shortage of **rare-earth magnets** alone can stop **EV** assembly lines.
- **Costly:** buying, storing and refreshing reserves takes **large public money**, and **price swings** add risk.
- **Only a bridge:** a stockpile buys **time**, not **long-term security**; **recycling** helps but depends on the **waste available**.

What should be done

- **Build domestic processing** from **oxides to finished components**.
- **Integrated magnets:** the Centre approved a **₹7,280-crore scheme** for **6,000 tonnes a year** of **integrated rare-earth permanent-magnet** capacity – from oxide to magnet.
- **Diversify imports** through partnerships and **stock processed materials**, not just ore.
- **Clear release rules:** which sectors get reserves, **how much and for how long**.
- **Bring in private firms** and promote **recycling** of batteries and e-waste; map which dependencies need **imports, alternate sources or home production**.
- **Value addition:** the **National Critical Mineral Mission (2025)**; India's list of **30 critical minerals (2023)**; **KABIL** acquires mineral assets abroad (e.g., **lithium blocks in Argentina**); India is in the **Minerals Security Partnership**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **rare earths, lithium, cobalt**, their **uses** and **where they are found**; and about **KABIL** and the **MMDR Act**.

PYQ – MAINS

- **Pattern:** Mains asks about **energy transition**, **supply-chain resilience** and **strategic minerals**.

FQ – PRELIMS (Possible Question)

- Consider: 1. Most of the world's rare-earth processing capacity is in China. 2. Rare-earth permanent magnets are used in EV motors and wind turbines. 3. A strategic stockpile alone ensures long-term supply security. Which are correct? **(Ans: 1 and 2 only)**

FQ – MAINS (Possible Question)

- "India's mineral security needs processing power, not just stockpiles." Discuss. (250 words)

GS-III | Agriculture – Animal Husbandry, Fisheries & Trade

8. Seafood and Buffalo Meat Lead India's Farm Exports

Syllabus: *GS-III: Animal husbandry and fisheries; agricultural marketing and exports; food processing; food security.*

> **CONTEXT**
>
> **Commerce Department data** for the **first half of 2026–27** show **marine products ($4.5 billion)** and **buffalo meat ($3.4 billion)** as India's **top farm exports**, ahead of **basmati rice** and other crops.



Illustration: refrigerated containers of shrimp and frozen meat being loaded on a ship beside a cold store and a fishing boat.

KEYWORDS DEFINED

Keyword	Meaning
Carabeef	**Buffalo meat** exported from India; India does not export beef from cows.
SPS measures	**Sanitary and phytosanitary** rules on **food safety and animal and plant health** that trading partners impose.
FMD	**Foot-and-mouth disease** – a highly contagious viral disease of cloven-hoofed animals that **blocks meat exports** to rich markets.
Vannamei shrimp	**Whiteleg shrimp** – the main farmed shrimp species behind India's seafood boom.
Cold chain	An unbroken system of **refrigerated storage and transport** from farm to buyer.

The numbers

- **Total farm exports** rose **4.7%** to **$27.1 billion** in **April–September 2026**.
- **Marine products** grew **13.2%** to **$4.5 billion**; **buffalo meat** jumped **53.8%** to **$3.4 billion**.
- **Seafood** hit a **record $8.4 billion in 2025–26**; buffalo meat's **average price** rose from about **$2,717 a tonne (2022–23)** to about **$4,368**.
- **Livestock and fisheries** now make up roughly **40% of farm GDP**, up from about **30% a decade ago**.

Why animal products are rising

- **Many markets:** buffalo meat goes to **Vietnam, Egypt, Malaysia, Indonesia and Saudi Arabia**; shrimp increasingly to the **EU, China and Japan** beyond the **US**.
- **Better quality:** only **registered modern abattoirs** meeting **SPS standards** may export; Indian carabeef is now sold as **lean, halal-certified meat**.
- **Aquaculture boom: vannamei** shrimp farms in **Andhra Pradesh, Odisha and Gujarat** with **disease-free hatcheries**.
- **Policy stability:** unlike **wheat, rice and sugar**, these items rarely face **sudden export bans or minimum export prices**.
- **Meanwhile crops wobble:** India became a **net cotton importer**; **edible oil and pulse duties** were cut and **sugar exports banned** after a weak monsoon.

Challenges

- **Strict SPS barriers: antibiotic residue** rejections of seafood in the **US and EU**.
- **FMD:** without **FMD-free status**, Indian meat is kept out of **North America and the EU**.
- **Environmental cost:** shrimp ponds and abattoirs cause **saline effluent, groundwater salinity and mangrove loss** (e.g., the **Godavari delta**).
- **Freight and conflict: West Asia tensions** push up **reefer container** rates on Gulf routes.
- **Value addition:** the **National Animal Disease Control Programme (2019)** aims to **control FMD and brucellosis through vaccination**; **PMMSY (2020)** funds fisheries; the **WOAH** (earlier OIE) certifies **disease-free zones**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **PMMSY**, **Blue Revolution**, **livestock diseases** and **India's top farm exports**.

PYQ – MAINS

- **Pattern:** Mains asks about **diversifying farm incomes** through **livestock and fisheries** and about **export competitiveness**.

FQ – PRELIMS (Possible Question)

- Consider: 1. Marine products were India's largest farm export in H1 2026–27. 2. India exports beef from cows. 3. Foot-and-mouth disease restricts Indian meat exports to Western markets. Which are correct? **(Ans: 1 and 3 only)**

FQ – MAINS (Possible Question)

- "Animal products are reshaping India's agricultural trade." Discuss the drivers and the challenges of sustaining this growth. (250 words)

GS-III | Indian Economy – Agricultural Export Institutions

9. Reforming APEDA: From Regulator to Export Think-Tank

Syllabus: *GS-III: Agricultural marketing and exports; food processing; issues of institutional reform.*

CONTEXT

Many of India's **high-value farm products** are not reaching their **export potential**, and experts say **APEDA** – the body in charge of farm and processed-food exports – must change from a **rule-keeper into a professional think-tank**.



Illustration: export boxes of mangoes, bananas, pomegranates, rice and organic produce at a port, with a testing lab and trade routes.

KEYWORDS DEFINED

Keyword	Meaning
APEDA	**Agricultural and Processed Food Products Export Development Authority** – set up under an **Act of 1985**, under the **Ministry of Commerce and Industry**.
NPOP	**National Programme for Organic Production** – India's organic certification system; **APEDA is its secretariat**.
Aflatoxin	A **toxic, cancer-causing chemical** made by moulds on groundnuts, maize and spices; buyers set **strict limits**.
Maximum residue limit (MRL)	The **highest legal level** of a pesticide or chemical allowed in food.
Geographical Indication (GI)	A tag showing a product's **quality comes from its place of origin** – e.g., **basmati**.

About APEDA

- Set up in **December 1985** under the **APEDA Act, 1985**; **headquarters New Delhi**; under the **Ministry of Commerce and Industry**.
- Handles **export promotion, quality, market and infrastructure development**, and **certification**.
- Serves as the **secretariat** to the **National Accreditation Board** that accredits certifiers under **NPOP**.

Problems

- **Thin budget** – about **₹80 crore a year** in most years since **2019–20**; **top posts vacant** for over **three years**.
- **Narrow basket: marine products, non-basmati rice, basmati and buffalo meat** were about **half** of farm exports worth **$53.72 billion in 2023–24**.
- **Organic credibility:** of nearly **20 lakh** organic farmers, only about **6 lakh** have been **verified**.
- **Rejections:** the **EU** turned back **1,200 containers of grapes (2010)** over **growth-regulator residues**; **Indonesia** curbed groundnuts over **aflatoxin**, and exports fell from **$300 million (2024)** to about **$73 million (Jan–Aug 2026)**.
- **GI setbacks** for **basmati** in **Australia, New Zealand and Kenya**; an **EU decision pending since 2018**.
- **Diaspora dependence:** produce sells mainly to **Indian communities abroad**, not mainstream chains like **Walmart or Tesco**.

Reforms suggested

- Become a **think-tank** for **research, market intelligence and strategy**, not just monitoring.
- **More money and staff**; fill **leadership posts** quickly.
- **Tighter organic certification** and **finish farmer verification**.
- **Train farmers and exporters** on **MRLs, aflatoxin and residue norms**.
- **Diversify** into **fruits, vegetables, pulses and processed food**; **brand Indian mangoes, bananas and pomegranates** in global retail.
- **Defend GIs** like basmati through **diplomatic and legal effort**.
- **Value addition:** the **Agriculture Export Policy (2018)** aimed to **double farm exports**; APEDA has helped register **GI-tagged** items and runs **traceability systems** like **GrapeNet** and **HortiNet**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **export bodies** (APEDA, MPEDA), **organic farming schemes** and **GI tags**.

PYQ – MAINS

- **Pattern:** Mains asks about **farm export competitiveness** and **quality standards**.

FQ – PRELIMS (Possible Question)

- APEDA functions under which ministry? (a) Agriculture and Farmers' Welfare (b) Food Processing Industries (c) Commerce and Industry (d) Consumer Affairs **(Ans: c)**

FQ – MAINS (Possible Question)

- Discuss the challenges APEDA faces in promoting India's agricultural exports and suggest measures to make it more effective. (250 words)

GS-III | Environment – Mine Closure & Land Restoration **UPDATE**

10. SUVIKALP: How the Mine Repurposing Tool Decides

Syllabus: *GS-III: Conservation, environmental degradation and restoration; mineral resources; e-governance tools.*

WHAT WE KNOW TILL NOW

- **SUVIKALP** is an **indigenous, data-driven tool** that **identifies and ranks new uses** for **closed and abandoned coal mines**.
- It was built by the **Coal Controller Organisation (CCO)**, an attached office of the **Ministry of Coal**, and won a **SKOCH Gold Award**.
- **PRISM** shows how to **implement** a chosen project, with **concept notes, sample DPRs and tender papers**.

WHAT TODAY ADDS

- How it **ranks options** – the **TOPSIS** method and a **five-stage** assessment.
- The **RECLAIM** framework for **community priorities**, and the tool now covering **lignite** mines too.

CONTEXT

The **Coal Controller Organisation** has rolled out **SUVIKALP**, its **digital decision-support platform** for turning **closed coal and lignite mines** into **productive, sustainable assets**.



Illustration: an old open-cast pit turned into a lake with floating solar panels, green terraces and a park, with ranked options on a tablet.

KEYWORDS DEFINED

Keyword	Meaning
TOPSIS	**Technique for Order of Preference by Similarity to Ideal Solution** – ranks options by how **close they are to the best** and **far from the worst**.
Multi-criteria decision-making	Choosing between options using **several criteria at once** – cost, ecology, jobs, feasibility.
Mine closure plan	A legally required plan to **close a mine safely** and **restore the land**.

Keyword	Meaning
Floating solar	**Solar panels on water bodies** – such as flooded mine pits.
Just transition	Moving away from coal **fairly**, with new jobs and skills for affected communities.

What it is

- An **indigenous digital platform** that **evaluates and ranks post-mining land uses** – **eco-parks, floating solar, afforestation, farming, industrial hubs**.
- **Developed by:** the **Coal Controller Organisation (CCO)**, **Ministry of Coal**.
- **Aim: scientific, transparent and participatory** closure that balances **technical feasibility, ecology and livelihoods**.

How it works

- **Site profile:** combines **geo-mining conditions, connectivity, environment and finance** for each site.
- **TOPSIS ranking:** scores options against **competing criteria**, reducing **personal bias**.
- **Five stages: factor selection → option generation → community priorities → suitability analysis → final ranking**.
- **RECLAIM:** brings **local preferences and livelihoods** – jobs, tourism, infrastructure – into the technical score.
- **PRISM:** gives **model DPRs, concept notes, case studies, tenders** and lists of **implementing agencies**.

Significance

- Turns **liabilities into assets** and supports **ecological restoration** and **regional development**.
- Promotes **evidence-based** decisions, the **circular economy** and **renewable energy** on degraded land.
- **Value addition:** India has **hundreds of closed or discontinued coal mines**; **Coal India** has developed **eco-parks and mine tourism** (e.g., in **Maharashtra and Madhya Pradesh**), and mine water is used for **drinking and irrigation** in coal belts.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **coal reserves, mine closure** and **land restoration** schemes.

PYQ – MAINS

- **Pattern:** Mains asks about **just transition** in coal regions and **rehabilitation of mining areas**.

FQ – PRELIMS (Possible Question)

- The TOPSIS method, used in the SUVIKALP tool, is a technique of: (a) satellite mapping (b) multi-criteria decision-making (c) carbon accounting (d) soil testing **(Ans: b)**

FQ – MAINS (Possible Question)

- "Closed mines can become engines of local growth." Discuss with reference to India's mine repurposing efforts. (150 words)

GS-III | Internal Security – Central Armed Police Forces

11. CRPF: A Year Without Operational Deaths, a First in 47 Years

Syllabus: *GS-III: Security forces and agencies and their mandate; left-wing extremism. GS-II: Welfare of personnel.*

CONTEXT

The **Central Reserve Police Force (CRPF)** has gone **12 straight months without an operational fatality** – the **first such year since January 1979**; its last operational death was in **late October 2025**.



Illustration: a patrol vehicle on a forest road, an evacuation helicopter, a first-aid kit and a helmet – safer operations and care for personnel.

KEYWORDS DEFINED

Keyword	Meaning
CAPF	**Central Armed Police Forces** under the **Ministry of Home Affairs** – CRPF, BSF, CISF, ITBP, SSB (and Assam Rifles is often grouped with them).
Casualty evacuation (CASEVAC)	Moving the **injured quickly** from the field to **medical care**.
Golden hour	The **first hour after injury**, when quick treatment most improves survival.
LWE	**Left-wing extremism** – Maoist insurgency in parts of central and eastern India.
Duty of care	An organisation's duty to **protect the safety and well-being** of its people.

About the CRPF

- India's **largest CAPF**, under the **Ministry of Home Affairs**.
- Raised in **1939** as the **Crown Representative's Police**; renamed **CRPF** under the **CRPF Act, 1949**.
- **Role:** helps States with **internal security, counter-insurgency, anti-Naxal operations, riot control and elections**.
- Specialised wings include **CoBRA** (jungle warfare against LWE) and the **Rapid Action Force (RAF)** for riots.

Why fatalities fell

- **Better equipment, training and mission planning** in high-risk areas.

- **Faster evacuation and treatment**, including quick access to **secondary and tertiary hospitals**.
- Continued **operations and breakthroughs** across major theatres **without losing personnel**.

Concerns that remain

- **Suicides and stress** among personnel point to a need for **mental-health care, counselling and welfare**.
- **Long deployments** away from family, and **leave and housing** issues.
- **Value addition:** the Centre has set **March 2026** as the target to **end LWE**; the number of **most-affected districts** has fallen sharply over the past decade *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **CAPFs and their ministries**, and about **specialised units** like CoBRA and RAF.

PYQ – MAINS

- **Pattern:** Mains asks about **LWE**, **police reforms** and the **welfare of security personnel**.

FQ – PRELIMS (Possible Question)

- Consider: 1. The CRPF functions under the Ministry of Defence. 2. It was raised in 1939 as the Crown Representative's Police. 3. CoBRA is a specialised CRPF unit for jungle warfare. Which are correct? **(Ans: 2 and 3 only)**

FQ – MAINS (Possible Question)

- "Zero operational deaths is a milestone; zero suicides should be the next." Discuss the welfare challenges of India's Central Armed Police Forces. (150 words)

GS-III | Environment – Wildlife & Disease (Species in News)

12. Markhor: Foot-and-Mouth Disease Threatens Kashmir's Wild Goats

Syllabus: *GS-III: Conservation – species and protected areas; wildlife–livestock interface. GS-I: Distribution of fauna.*

CONTEXT

Foot-and-mouth disease (FMD) outbreaks among **livestock in Kashmir** have raised fears that the virus could spread to **wild animals** such as the **markhor and the hangul**.



Markhor, the world's largest wild goat, known for its corkscrew horns (photographed in a zoo).
Photo: NasserHalaweh, Wikimedia Commons, CC BY-SA 4.0

KEYWORDS DEFINED

Keyword	Meaning
Markhor	*Capra falconeri* – the **largest wild goat** in the world, with **spiral horns**; the **national animal of Pakistan**.
Hangul	The **Kashmir stag** (*Cervus hanglu hanglu*), found mainly in **Dachigam National Park**.
Spillover	When a disease **jumps from one species to another** – e.g., from cattle to wildlife.
FMD	A **viral disease of cloven-hoofed animals** causing **blisters in the mouth and feet**; rarely affects humans.
Near Threatened	An **IUCN** category for species **close to qualifying as threatened**.

About the markhor

- **Scientific name:** *Capra falconeri*; the **world's largest wild goat**.
- **Looks:** males have **corkscrew horns up to about 160 cm**, a **long beard** and thick **brown-grey coat** for camouflage; males weigh about **80–110 kg**, females about **32–50 kg**.
- **Habitat:** steep mountains at about **600–3,600 m** in **Pakistan, Afghanistan, India and Tajikistan**; in India, mainly the **Pir Panjal and Kazinag** ranges of **J&K**.

- **Diet and habits:** grazes grass and leaves, browses woody plants in winter; **females and young** live in herds, **males** alone or in small groups.

Threats and status

- **Poaching for horns**, **habitat loss** and **competition with livestock** for grazing.
- **IUCN: Near Threatened**; in India listed in **Schedule I** of the **Wildlife (Protection) Act, 1972** *(as amended)*.
- **New risk: FMD** from domestic animals sharing pastures.

Why disease spillover matters

- Small, isolated wild populations can **crash** after an outbreak.
- **One Health:** human, livestock and wildlife health are **linked**; **vaccinating livestock** near protected areas protects wildlife too.
- **Value addition:** the **hangul** is **Critically Endangered** and survives mainly in **Dachigam**; FMD control is part of the **National Animal Disease Control Programme**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **Himalayan fauna**, their **IUCN status** and **protected areas**.

PYQ – MAINS

- **Pattern:** Mains asks about **human–wildlife–livestock conflict** and **One Health**.

FQ – PRELIMS (Possible Question)

- Consider: 1. The markhor is the largest wild goat in the world. 2. Its IUCN status is Critically Endangered. 3. Foot-and-mouth disease can spread from livestock to wild ungulates. Which are correct? **(Ans: 1 and 3 only)**

FQ – MAINS (Possible Question)

- "Wildlife conservation now needs veterinary security." Discuss with reference to disease spillover from livestock. (150 words)

GS-III | Environment – Biodiversity (New Species)

13. Bupleurum oslaense: A New Herb From Uttarakhand

Syllabus: *GS-III: Conservation – biodiversity, endemic species and protected areas.*

CONTEXT

Botanists of the **Botanical Survey of India (BSI)** have described a **new plant species**, *__Bupleurum oslaense__, *from the Western Himalayas in Uttarakhand**.



Illustration: a slender Himalayan herb with narrow and broad leaves and small flower clusters in a conifer forest.

KEYWORDS DEFINED

Keyword	Meaning
Botanical Survey of India (BSI)	The **apex plant-survey body** under the **Ministry of Environment, Forest and Climate Change**, set up in **1890**, HQ **Kolkata**.
Perennial	A plant that **lives for more than two years**.
Apiaceae	The **carrot family** – includes **coriander, cumin, fennel** and many herbs.
Bract	A **leaf-like part** at the base of a flower or flower cluster.
Critically Endangered	The **IUCN category** facing an **extremely high risk** of extinction in the wild.

About the species

- A **perennial herb** of **temperate forests** in the **Western Himalayas**, family **Apiaceae**.
- **Found near Osla village**, Uttarakhand, at about **2,550 m**, inside **Govind Pashu Vihar National Park and Sanctuary**; **named after Osla**.
- Closely resembles *__Bupleurum petiolulatum__, *a species from China**.

Key features

- **Height:** up to about **80 cm**; a **sturdy** herb.
- **Leaves: narrow lower leaves** and **broader upper leaves** with **red-tinged edges**.
- **Flowers:** clusters of **10–20 flowers** with **3–5 larger bracts**; **fruits broadly ovoid** with slightly **winged ribs**.

Status and threats

- Assessed as **Critically Endangered** (IUCN criteria) because of its **tiny range and small population**.
- **Value addition: Govind Pashu Vihar** in **Uttarkashi** lies in the **Tons valley**; it is known for the **Har-ki-Dun** valley and as a home of the **snow leopard** and the **bearded vulture**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **new species**, their **families** and the **protected areas** where they were found.

PYQ – MAINS

- **Pattern:** Mains asks about **Himalayan biodiversity** and **taxonomy's role in conservation**.

FQ – PRELIMS (Possible Question)

- Bupleurum oslaense, recently described, was found in: (a) Great Himalayan National Park (b) Govind Pashu Vihar National Park (c) Valley of Flowers National Park (d) Dachigam National Park **(Ans: b)**

FQ – MAINS (Possible Question)

- "Every new species described is also a species at risk." Discuss with reference to narrow-range Himalayan endemics. (150 words)

GS-III | Environment – Species & Wetlands (Species in News)

14. Saltwater Crocodile Deaths in Bhitarkanika

Syllabus: *GS-III: Conservation – species, wetlands and protected areas; environmental impact. GS-I: Coastal ecosystems.*

CONTEXT

Within **27 days**, **five saltwater crocodiles** were found dead in and near **Bhitarkanika National Park** (Kendrapara, Odisha), taking **unnatural crocodile deaths** to **22 in six years** and **44 in 14 years**.



A saltwater crocodile slipping into the water at Bhitarkanika National Park, Odisha.
Photo: Souvik221917, Wikimedia Commons, CC BY-SA 4.0

KEYWORDS DEFINED

Keyword	Meaning
Saltwater crocodile	*Crocodylus porosus* – the **largest living reptile**, found in **estuaries and mangroves** of the Indo-Pacific.
Estuary	Where a **river meets the sea** – water is **brackish** (part salt, part fresh).
Salt glands	Glands on the crocodile's **tongue** that **remove excess salt**, letting it live in sea water.
Ramsar site	A **wetland of international importance** under the **Ramsar Convention (1971)**.
Apex predator	An animal at the **top of the food chain**, with no natural predators as an adult.

About the saltwater crocodile

- The **largest living reptile** and **largest crocodilian**; adult males reach about **6–7 m** and well over **1,000 kg**; females are much smaller (about **2.5–3 m**).
- **Strongest measured bite** of any animal – around **3,700 psi**.
- **Habitat:** mangroves, estuaries, deltas, lagoons; can **swim long distances at sea**; in India – **Odisha (Bhitarkanika), the Sundarbans and the Andaman & Nicobar Islands**.
- **Care of young:** the female builds a **mound nest**, lays about **40–90 eggs**, **guards it for about three months** and carries hatchlings to water in her **jaws**.

About Bhitarkanika

- A **national park (about 145 sq km)** inside the larger **Bhitarkanika Wildlife Sanctuary (about 672 sq km)** in **Kendrapara, Odisha**; a **Ramsar site since 2002**.
- **India's second-largest mangrove forest** after the **Sundarbans**, in the delta of the **Brahmani, Baitarani, Dhamra and Pathsala** rivers.
- Home to **India's largest wild saltwater crocodile population** – up from **95 in 1975** to about **1,858 in 2026** *(as cited)*.
- Borders the **Gahirmatha Marine Sanctuary**, a major **Olive Ridley** nesting beach; its mangroves **shield the coast from cyclones**.

Why the deaths matter

- **Rising human–crocodile conflict** as both numbers and **fishing pressure** grow.
- Likely causes include **fishing nets, poaching, fights and habitat stress** – each death needs a **post-mortem and probe**.
- **Value addition:** India's **crocodile conservation project (1975)**, with **FAO–UNDP** help, began at **Bhitarkanika**; the species is in **Schedule I** of the **Wildlife (Protection) Act** and **IUCN: Least Concern**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **crocodilians of India** (gharial, mugger, saltwater), their **habitats** and **IUCN status**.

PYQ – MAINS

- **Pattern:** Mains asks about **mangrove conservation** and **human–wildlife conflict**.

FQ – PRELIMS (Possible Question)

- Consider: 1. The saltwater crocodile is the largest living reptile. 2. Bhitarkanika has India's largest mangrove forest. 3. Bhitarkanika is a Ramsar site. Which are correct? **(Ans: 1 and 3 only)**

FQ – MAINS (Possible Question)

- "Conservation success can create new conflicts." Discuss with reference to saltwater crocodiles in Bhitarkanika. (150 words)

GS-III | Science & Technology – Space Science

15. NASA's Swift Observatory Falls Back to Earth After 22 Years

Syllabus: *GS-III: Science and technology – space; awareness in astronomy.*

CONTEXT

NASA's Neil Gehrels Swift Observatory, a **1.5-tonne** space telescope, is making an **uncontrolled re-entry** into Earth's atmosphere after **22 years** of work hunting **gamma-ray bursts**.

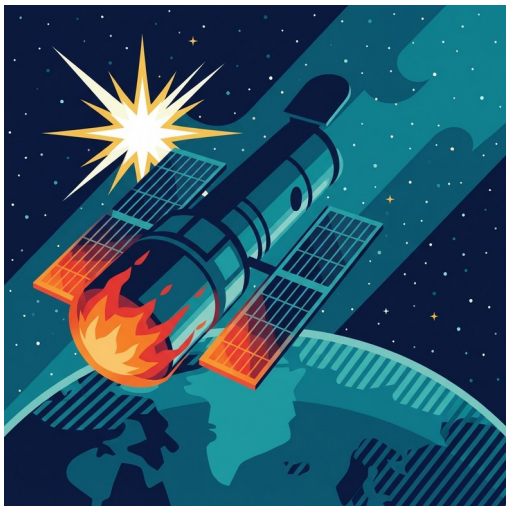


Illustration: a space telescope beginning to burn up as it re-enters the atmosphere, with a distant gamma-ray burst.

KEYWORDS DEFINED

Keyword	Meaning
Gamma-ray burst (GRB)	The **brightest explosions** in the universe – from **massive stars collapsing** or **neutron stars merging**.
Afterglow	The **fading glow** in X-ray, ultraviolet and visible light that follows a GRB.
Slewing	**Turning a telescope quickly** to point at a new target.
Uncontrolled re-entry	When a satellite **falls back without guided steering**; most of it **burns up**.
Multi-messenger astronomy	Studying an event through **different signals** – light, **gravitational waves**, neutrinos.

About Swift

- A **multi-wavelength space observatory** for **high-energy, short-lived events**, built to **detect GRBs on its own** and **turn its telescopes within seconds**.
- **Launched on 20 November 2004** on a **Delta II** rocket under NASA's **MIDEX** programme; **renamed in 2018** after its principal investigator **Neil Gehrels** (died 2017).
- Operated by **NASA Goddard**, in partnership with **Penn State** and with support from the **British and Italian** space agencies.
- Designed for **two years**, it worked for **over two decades** until its **orbit decayed**.

Instruments

- **Burst Alert Telescope (BAT):** spots **hard X-rays and soft gamma rays (15–150 keV)** across about **one-sixth of the sky**.
- **X-Ray Telescope (XRT):** pinpoints afterglows (**0.2–10 keV**).
- **UV/Optical Telescope (UVOT):** a **30-cm** telescope measuring early **brightness and colour**.
- Can **re-point in about 20–75 seconds** using reaction wheels.

Achievements

- Detected **over 2,000 GRBs**, revealing **stellar collapses and compact-object mergers**.
- Saw **GRB 090423** at **redshift about 8.2** – light from about **630 million years after the Big Bang**.
- Helped follow up the **neutron-star merger GW170817** – a landmark of **multi-messenger astronomy**; sends rapid alerts via the **General Coordinates Network**.
- **Value addition:** India's **AstroSat (2015)** has a **CZT Imager** that also detects GRBs; **XPoSat (2024)** studies **X-ray polarisation**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **space telescopes**, **black holes, neutron stars, gravitational waves** and India's **AstroSat**.

PYQ – MAINS

- **Pattern:** Mains asks about **space science missions** and **space debris**.

FQ – PRELIMS (Possible Question)

- Consider: 1. Swift was designed mainly to study gamma-ray bursts. 2. It was launched by ISRO. 3. It helped observe the neutron-star merger GW170817. Which are correct? **(Ans: 1 and 3 only)**

FQ – MAINS (Possible Question)

- "Old satellites are falling, new ones are rising." Discuss the problem of space debris and uncontrolled re-entries. (150 words)

GS-III | Science & Technology – Artificial Intelligence

16. EKA-1 and Dhīr: IndiaAI's New Home-Grown Foundation Models

Syllabus: *GS-III: Science and technology – developments in IT and AI; indigenisation of technology.*

> **CONTEXT**
> The **IndiaAI Mission** has released two sets of **indigenous foundation models** – **EKA-1** by **Soket Labs** and **Dhīr 2B and Dhīr 5B** by **ZenteiQ** – under its **Innovation Centre** pillar.

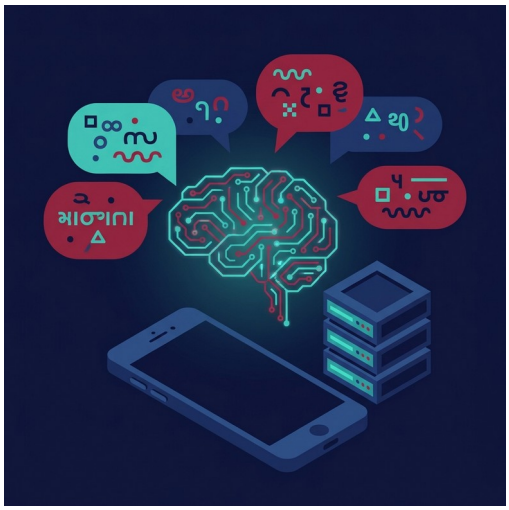


Illustration: a glowing AI brain above a phone and a server rack, with speech bubbles in many Indian scripts.

KEYWORDS DEFINED

Keyword	Meaning
Foundation model	A **large AI model trained on vast data** that can be **adapted to many tasks**.
Parameters	The **internal settings** an AI model learns; more parameters usually mean **more capacity**.
Tokens	**Pieces of text** (words or parts of words) used to train and run language models.
Sparse model	A model that **uses only part of its network for each input**, saving computing power.
On-device AI	AI that **runs on a phone or laptop itself**, without sending data to the cloud.

EKA-1 (Soket Labs)

- An **open, sparse hybrid language model** with about **26.3 billion parameters**, trained on roughly **7.5 trillion tokens**.
- Works in **all 22 scheduled Indian languages** plus **English**; built for **maths, coding, reasoning** and **agentic** tasks.
- Tailored for India's **banking, financial services and cyber-security** needs.

Dhīr 2B and 5B (ZenteiQ)

- **Open-source** small models (about **2 billion and 5 billion** parameters) that can run as **secure, on-premise assistants** in **English and 10 Indian languages**.

- Run on **GPUs and TPUs**, and are light enough to work **on mobile phones without the cloud**.
- Useful for **document-based automation and knowledge search** in **defence, government, industry and education** – even in **low-connectivity or sensitive settings**.

Why it matters

- **Sovereign AI:** models trained for **Indian languages and needs** reduce **dependence on foreign platforms**.
- **Privacy and security: on-device and on-premise** use keeps **data inside India** and inside institutions.
- **Value addition:** the **IndiaAI Mission (2024)** has an outlay of about **₹10,372 crore** with seven pillars, including **compute capacity**, **datasets (AIKosh)**, **foundation models** and **skilling**; **MeitY** is the nodal ministry.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **AI terms**, **IndiaAI Mission**, **Bhashini** and other **digital public goods**.

PYQ – MAINS

- **Pattern:** Mains asks about **AI for governance**, **data sovereignty** and **ethical AI**.

FQ – PRELIMS (Possible Question)

- Consider: 1. EKA-1 supports all 22 scheduled Indian languages. 2. Dhīr models need a cloud connection to run. 3. The IndiaAI Mission comes under MeitY. Which are correct? **(Ans: 1 and 3 only)**

FQ – MAINS (Possible Question)

- "Sovereign AI is as much about languages as about chips." Discuss with reference to India's indigenous foundation models. (150 words)

GENERAL STUDIES PAPER IV

[GS-IV] Ethics, Integrity & Aptitude

GS-IV | Ethics, Integrity & Aptitude – Synthesis

17. Ethics Synthesis: Duty of Care, Fairness and Stewardship

Syllabus: *GS-IV: Ethics in public administration; accountability; compassion; environmental ethics; integrity; case studies.*

CONTEXT

Today's topics meet on one ground: **those with power must care for those who depend on them**. A **security force** caring for its men, a **fair deal with a neighbour** over a river, **honest certification** of organic food, and **wildlife held in trust** all test **duty of care, fairness and stewardship**.

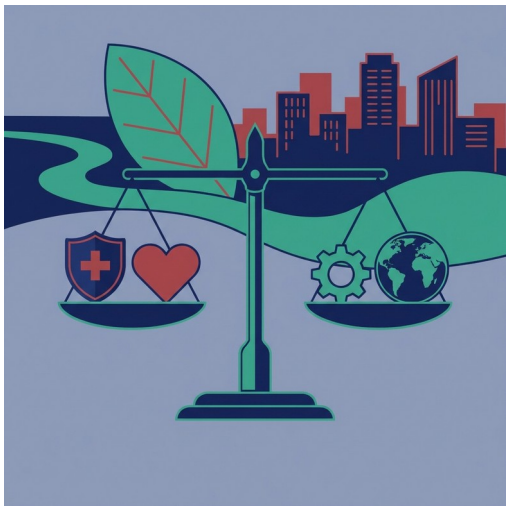


Illustration: a balance weighing care and safety against efficiency and the wider world.

KEYWORDS DEFINED

Keyword	Meaning
Duty of care	A duty to **protect the safety and well-being** of those who **depend on you**.
Distributive justice	**Fair sharing** of benefits and burdens – e.g., of **river water**.
Integrity	Acting **honestly and consistently** with values, even when no one is watching.
Stewardship	Holding resources and nature **in trust** for others and future generations.
Good neighbourliness	Respecting a neighbour's **needs and dignity** in shared affairs.

The ethical thread

- **Duty of care:** the **CRPF's** year without operational deaths shows **planning and quick medical help save lives** – but **suicides and stress** show care must include the **mind**.
- **Fairness:** the **Ganga treaty** asks how to share water **justly** in **dry months and a changing climate**.
- **Integrity:** only **6 of 20 lakh** organic farmers verified – **trust in certification** depends on **honest checks**.
- **Stewardship: crocodiles, markhor** and a **new Himalayan herb** remind us that nature is **held in trust**.

- **Responsibility with technology: AI models** and **critical minerals** bring power that must serve **people and privacy**.

Values at stake

- **Compassion, duty of care, fairness, integrity, accountability, stewardship, prudence, empathy.**

Case study 1: The stressed jawan

- **Situation:** You command a **CRPF battalion** in an LWE area. A **constable** who has not had leave for months seems **withdrawn and irritable**; a colleague reports he talked about 'ending it all'.
- **Issues: mental health vs operational pressure**; stigma; duty of care.
- **Options:** (a) ignore it – everyone is under stress; (b) disarm him and send him for disciplinary action; (c) **talk to him privately**, arrange **counselling and medical help**, **grant leave**, keep his **weapon safely** for now, and **train officers** to spot warning signs.
- **Preferred course: (c)** – **compassion with prudence**; care for the person **is** operational readiness.

Case study 2: The organic certificate

- **Situation:** As an **inspector of a certification body**, you find a big exporter's 'organic' farms using **chemical pesticides**. The firm offers a **'consultancy fee'** and warns that a rejection will **cost hundreds of jobs**.
- **Issues: integrity vs pressure**; consumer trust; India's export reputation.
- **Options:** (a) certify and take the money; (b) quietly delay the report; (c) **refuse the bribe**, **record the findings**, **report the offer**, and suggest a **conversion period** so the firm can become genuinely organic.
- **Preferred course: (c)** – one false certificate can **sink trust in all Indian organic exports**.

Case study 3: The crocodile and the fishers

- **Situation:** As **forest officer** at Bhitarkanika, you face **angry fishers** after a crocodile attack, demanding the animals be **removed**; meanwhile **nets are killing crocodiles**.
- **Issues: livelihoods vs conservation**; safety; trust.
- **Options:** (a) cull problem animals; (b) ban fishing entirely; (c) **compensate victims quickly**, mark **safe bathing and fishing zones**, give **crocodile-safe nets**, and **involve fishers** in monitoring.
- **Preferred course: (c)** – **co-existence** built on **fairness and participation**.

Thinkers to quote (paraphrased)

- **Mahatma Gandhi:** the **greatness of a nation** is seen in how it treats the **weakest**.
- **John Rawls:** fair rules are those we would choose **without knowing our own place** – the 'veil of ignorance'.
- **Immanuel Kant:** act so that your rule could be a **universal law** – no exception for a bribe.
- **Aldo Leopold:** a thing is right when it **protects the integrity of the living community**.
- **Swami Vivekananda: serve people as you would serve God**.

PYQ – PRELIMS

- **Not applicable** – GS-IV is Mains-only.

PYQ – MAINS

- **Pattern:** GS-IV often asks about **compassion towards the weaker sections**, **integrity under pressure**, **environmental ethics** and **case studies** on conflicting duties.

FQ – PRELIMS (Possible Question)

- **Not applicable** – GS-IV is Mains-only.

FQ – MAINS (Possible Question)

- "Duty of care is the moral core of leadership." Discuss with examples from public service. (150 words)
- Case study: As an organic-certification inspector, how would you respond to a bribe offered with a threat of job losses? (250 words)